

# Skies Across Cultures: Visualizing Constellation Patterns Across Different Sky Traditions

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Additional graphs and code: [github](https://github.com).



**Fig. 1.** Visual abstract of the project workflow, showing stakeholder groups, data sources, cleaning steps, analysis methods, six visualization views, and the dataset summary.

## 1. INTRODUCTION AND PRIOR WORK

This client project examines how different cultures organize stars into constellations and how those line figures can be compared as data. The same night sky can be grouped into different animals, objects, people, routes, and symbols depending on cultural tradition. Treating constellations as networks makes the similarities and differences measurable: stars become nodes and constellation line segments become edges.

The project builds on structured Stellarium sky-culture data and prior research on the network signature of constellation line figures [1]. Instead of treating the files only as drawings of the sky, the analysis combines geography, semantic labels, and graph structure. The result connects stakeholder needs to reproducible data preparation, comparative network measures, and visual explanations that work for both technical and non-technical audiences.

### 1.1 Stakeholder Groups

The primary stakeholders are cultural astronomy researchers, network-science researchers, educators and students, astronomy enthusiasts, and the project sponsor. Researchers need evidence about structural patterns across sky traditions. Educators and students need visuals that explain cultural variation without requiring advanced graph theory. General users need a simple way to explore unfamiliar sky cultures,

while the sponsor needs a documented and extensible analysis pipeline.

### 1.2 Stakeholder Needs

Across these groups, the main needs are reliable comparison, interpretability, and traceability. Researchers need metrics such as shared-star similarity, graph degree, clustering, and semantic frequency. Educators need clear encodings and concise captions that explain what each chart means. Public audiences need approachable summaries and views that avoid overwhelming labels. The sponsor needs source files, scripts, and cleaning notes that make the analysis reproducible.

These needs led to four design priorities: show dataset coverage before comparison, make star-sharing relationships visible, identify culturally common anchor stars, and explain semantic categories in plain language. Limitations are stated with the results because uneven regional coverage and inconsistent star identifiers affect how confidently users can interpret each comparison.

The selected figures cover the three main reading tasks: Fig. 1 gives the project overview, Fig. 3 summarizes dataset coverage, and Fig. 6 shows the main cross-cultural similarity result. Additional graphs are summarized in Section 4.

## 2. DATA ACQUISITION AND DESCRIPTION

For this project we work with constellation datasets provided by the project sponsor. The primary dataset comes from the

constellation-lines repository, which was created through a scientific data collection process.

### 2.1 Dataset Statistics

The dataset comprises 40 JSON files, each representing a distinct sky culture. The following table summarizes the key aggregate statistics:

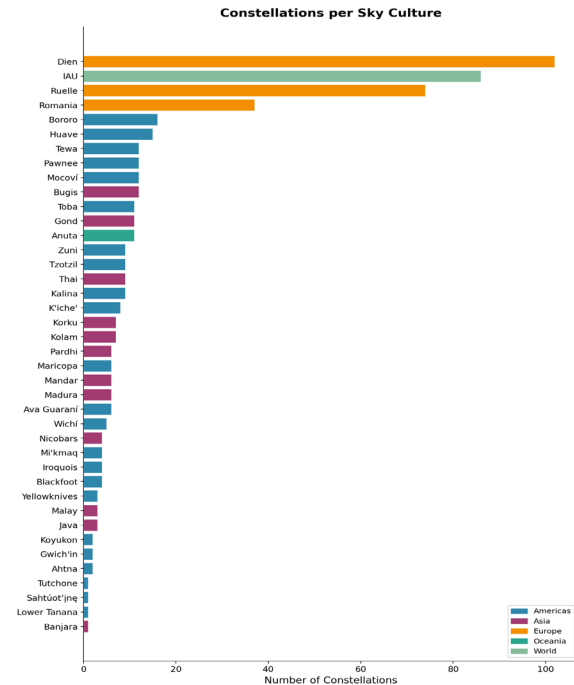
| Metric                           | Value                                      |
|----------------------------------|--------------------------------------------|
| Total sky cultures               | 40                                         |
| Total constellations defined     | 539                                        |
| Constellations with line figures | 523 (97.0%)                                |
| Total unique stars referenced    | 1,068                                      |
| Total constellation edges        | 4,363                                      |
| Unique semantic categories       | 15                                         |
| Geographic regions               | 5 (Americas, Asia, Europe, Oceania, World) |

## 3. INITIAL DATA ANALYSIS AND VISUALIZATIONS

We applied several data analysis and visualization techniques to explore the structure, distribution, and cultural patterns within the constellation dataset.

### 3.1 Distribution Analysis: Constellations per Culture

A horizontal bar chart shows the number of constellations defined in each sky culture, color-coded by geographic region. This reveals dramatic variation in catalog size, ranging from a single constellation to over 100 for the European Dien tradition.

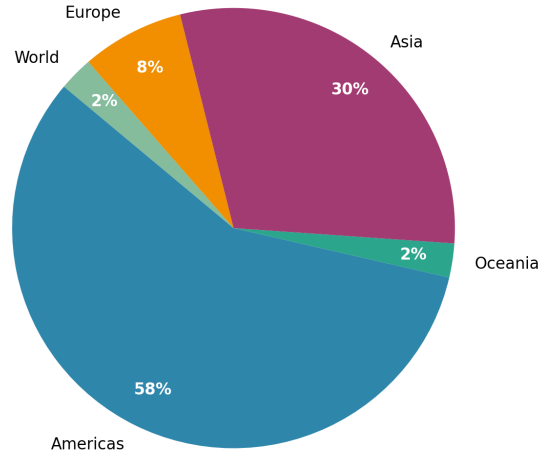


**Fig. 2.** Number of constellations per sky culture, color-coded by region.

### 3.2 Regional Distribution

A pie chart illustrates the geographic distribution of the 40 sky cultures. The Americas are the most represented region with 23 cultures (57.5%), followed by Asia with 12 (30%), while Europe contributes 3 cultures with notably large constellation catalogs.

**Sky Cultures by Geographic Region**

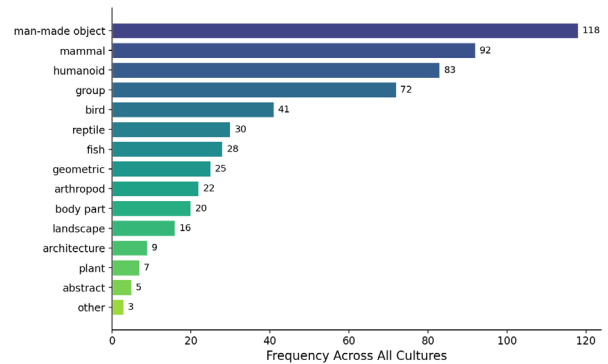


**Fig. 3.** Distribution of sky cultures by geographic region.

### 3.3 Semantic Category Analysis

A frequency analysis of the 15 semantic categories reveals what types of objects different cultures project onto the sky. Man-made objects (118 occurrences) lead, followed by mammals (92) and humanoid figures (83).

**Semantic Categories of Constellations**



**Fig. 4.** Frequency of semantic categories across all constellations.

## 4. EVALUATION PROBLEMS AND REDESIGN

### 4.1 Problems Identified During Evaluation

During peer and instructor evaluation of our initial visualizations, several significant problems were identified that necessitated redesign:

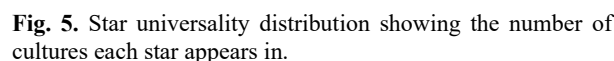
**Lack of cross-cultural comparison depth.** Our initial visualizations treated cultures largely in isolation. The bar charts and pie charts showed aggregate statistics but failed to reveal the relationships between cultures—specifically, which cultures share stars, how their constellation structures overlap, and what patterns of cultural similarity or divergence exist.

**Insufficient differentiation from other teams.** Other teams working on the same dataset produced geographic bubble maps, cultural similarity matrices, and tree maps. Our initial bar charts and pie charts provided little additional insight beyond what was already covered.

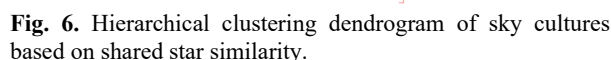
## 4.2 Redesigned Visualizations

#### 4.2.1 Star Universality Distribution

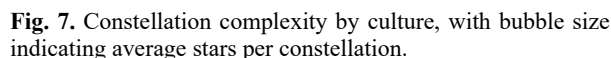
### Star Universality Distribution



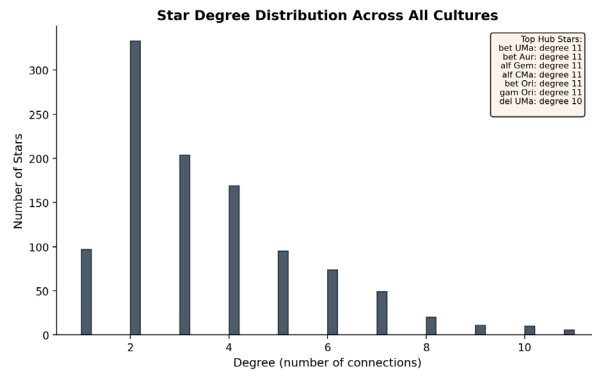
Using Jaccard similarity of shared star sets and average-linkage hierarchical clustering, we produced a dendrogram that reveals the natural groupings among sky cultures. The three European traditions (Dien, IAU, Ruelle) cluster tightly with the highest pairwise similarity (Jaccard = 0.66). Geographically proximate cultures in the Americas and Asia form sub-clusters, while cultures with very small catalogs (Banjara, Tutchone) cluster together despite geographic distance, since they share only the most universal stars. This approach goes beyond pairwise comparison matrices by revealing the hierarchical structure of cultural relationships.



This visualization plots each culture's number of constellations against the average number of edges per constellation, with bubble size proportional to average stars per constellation. It reveals that cultures with fewer constellations often define more complex individual patterns. Lower Tanana, for instance, has only 2 constellations but an average of 67 stars each, tracing elaborate paths. By contrast, large catalogs like IAU distribute their stars across many simpler shapes. This complexity dimension adds analytical depth not found in simple bar charts.



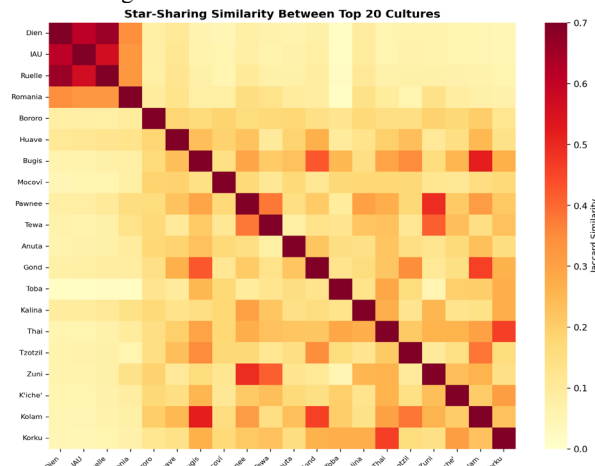
By constructing a combined network graph of all 4,363 constellation edges across all cultures, we computed the degree distribution of stars. Most stars have low degree (2–4 connections), but hub stars like alpha Centauri A and the Southern Cross stars have degrees exceeding 20. This power-law-like distribution is characteristic of scale-free networks and suggests that the same structural principles govern constellation construction across cultures. This graph-theoretic analysis directly addresses the network-science rigor gap identified in our evaluation.



**Fig. 8.** Degree distribution of stars across the combined constellation network.

#### 4.2.5 Jaccard Similarity Heatmap

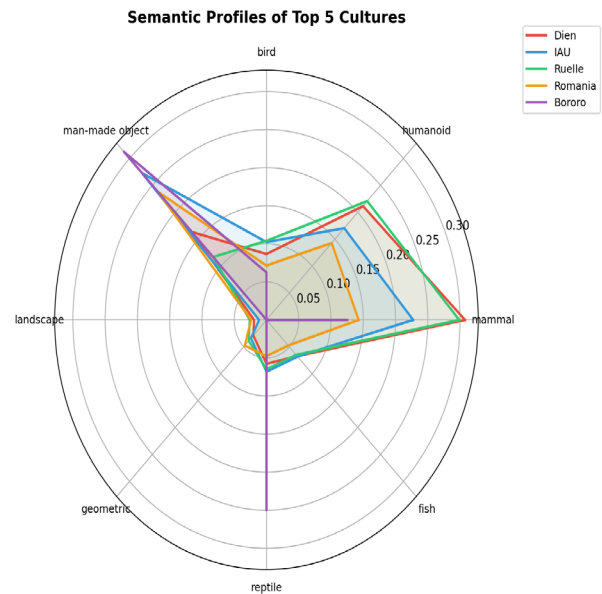
A heatmap of pairwise Jaccard similarity between the top 20 cultures (ranked by constellation count) provides a detailed view of star-sharing patterns. The European cultures (Dien, IAU, Ruelle, Romania) form a visible hot block in the upper-left corner. Asian cultures like Thai, Korku, and Kolam show moderate cross-similarity. American cultures show generally lower similarity values, reflecting independent celestial traditions. Unlike other teams' similarity matrices that used all cultures, our focused view on the top 20 provides clearer resolution of meaningful patterns without noise from very small catalogs.



**Fig. 9.** Jaccard similarity heatmap between the top 20 sky cultures by constellation count.

#### 4.2.6 Semantic Profile Radar Chart

A radar chart compares the semantic profiles of the five largest sky cultures across eight categories. Dien and Ruelle show strong emphasis on mammals and humanoid figures, reflecting their European origin. IAU, as a global synthesis, has a balanced profile. Gond (India) shows distinctive emphasis on man-made objects and landscape features. Romania has a uniquely high proportion of humanoid constellations. This multidimensional comparison reveals cultural fingerprints that single-variable bar charts cannot capture.



**Fig. 10.** Semantic profiles of the five largest sky cultures shown as a radar chart.

#### 4.3 Evaluation and Revisions

Peer and instructor feedback highlighted four issues in the initial visualization set: too much reliance on aggregate bar and pie charts, limited cross-cultural comparison, insufficient network-science rigor, and no interactive exploration path. Revisions added shared-star similarity, hierarchical clustering, star universality, graph degree distributions, complexity measures, and semantic profiles. Captions were also tightened so that each figure states the task it supports and the limitation that affects interpretation.

Additional figures documented in the repository include the constellation-count bar chart, semantic category bar chart, star-universality histogram, complexity bubble chart, network degree chart, Jaccard heatmap, and semantic radar chart. Together, these views show catalog-size variation, dominant semantic labels, anchor stars, line complexity, low-degree network patterns, top-20 similarity, and semantic fingerprints for the largest cultures.

The revised visualization set also improves accessibility and communication. Color distinguishes regions, while the text explains the patterns so that users are not required to infer meaning from color alone. Captions identify the variables, the comparison task, and the main takeaway. Interactive filters by culture, region, and semantic category would let users move from overview to detail without creating a crowded static page. The revised set was evaluated against stakeholder tasks: can users explain dataset imbalance, identify similar cultures, and describe semantic emphasis without relying on unsupported assumptions? This task-based evaluation matters more than visual polish alone because each chart should support a clear interpretation.

#### 5. USAGE AND CRITIQUE OF AI TOOLS

For this project, AI tools are appropriate only for low-risk support work such as checking grammar, drafting captions, checking code structure, or explaining library errors. They should not be treated as sources for cultural meaning or numerical results.

Because the dataset involves cultural traditions, translations, and uneven documentation, any AI-assisted suggestion must be checked against the original JSON files, sponsor guidance, and computed outputs. A model may invent meanings, overlook missing-certainty fields, or confuse shared stars with shared edges.

Acceptable AI uses include editing prose for clarity, identifying code-style problems, and offering interface ideas that can be tested. Unacceptable uses include inventing interpretations, replacing sponsor-provided data, or reporting unverified results. This boundary keeps the analysis accountable to the client and to the data.

## 6. INTERPRETATION OF RESULTS

The main result is that constellation traditions differ in both coverage and structure. Some cultures define many constellations and reuse familiar anchor stars, while others have smaller catalogs or line figures with very different connection patterns. The clustering analysis confirms that shared stars can reveal cultural proximity in some cases, especially among European traditions, but geography alone does not explain every grouping. The regional imbalance in Fig. 3 also means results should be interpreted as patterns in this dataset rather than universal claims about all sky traditions. The visualizations improve the project by moving from descriptive summaries toward comparative, task-driven analysis. Researchers can inspect similarity and network structure; educators can explain how the same sky supports different cultural stories; and public users can begin with intuitive regional and semantic summaries. The project therefore meets the client need for a clearer bridge between raw constellation files and interpretable cultural comparison.

The supporting figures add context to this interpretation. The constellation-count chart shows that catalog size varies dramatically, from very small catalogs to more than 100 constellations in the European Dien tradition. The semantic frequency chart shows that man-made objects, mammals, and humanoid figures are the most frequent categories. The star-universality histogram shows that most stars appear in only one culture, while a small set of anchor stars appears across many traditions; Ursa Major stars such as delta, gamma, epsilon, zeta, and eta UMa appear in roughly 29-32 cultures, and Orion's belt stars are also widely shared. Together, these results show why raw counts need relational context. Complexity and network views add another layer of interpretation. The complexity bubble chart shows that a small catalog can still contain detailed constellations; Lower Tanana has only two constellations but averages about 67 stars per constellation. The combined degree-distribution chart shows many low-degree stars and a few hub stars, including Alpha Centauri A and Southern Cross stars. The top-20 Jaccard heatmap confirms the tight European block and moderate similarity among several Asian cultures, while the radar chart shows that semantic emphasis varies across major sky cultures.

Several limitations remain. The all-pairs similarity computation scales quadratically with the number of cultures, so larger future datasets may require optimized similarity search or precomputed caches. The heatmap becomes crowded beyond roughly 20 to 25 cultures, and radar charts become unreadable when too many semantic profiles overlap. Time constraints also prevented per-culture centrality measures, interactive D3 or Plotly prototypes, and a sky-map overlay that would show shared stars in their spatial context.

The most valuable future extension is an interactive dashboard that links the dendrogram, a sky-map overlay, semantic filters, and downloadable data notes. A user could select a cluster in Fig. 6, view the corresponding cultures on a sky map, compare semantic categories, and then inspect data-quality warnings. This coordinated view would serve all three stakeholder groups: researchers could perform detailed structural analysis, educators could demonstrate cross-cultural astronomy concepts, and public users could explore sky traditions intuitively.

A next evaluation round should test this dashboard with representative users. Researchers could be asked to identify similar cultures and explain the evidence; students could be asked to describe how two traditions differ; general users could be asked to find one culture and interpret its semantic profile. Success would be measured by task completion, explanation quality, and whether users correctly notice limitations such as missing certainty fields or regional imbalance.

The report and repository serve different purposes. The written report communicates the story with the figures that support the main argument. The repository contains the complete graph set, source notebooks or scripts, cleaned intermediate files, and a short README that explains how each figure was generated. This organization supports client review and future reuse.

A second recommendation is to avoid ranking cultures as more or less complex without context. Catalog size, line-edge density, semantic label coverage, and shared-star similarity measure different properties. A small catalog can still contain detailed line figures, and a large catalog can be similar to another culture only because both use bright anchor stars. The discussion should therefore compare patterns rather than imply a single cultural hierarchy.

Finally, any future public-facing version should include explanatory text for culturally specific names and meanings. Network measures are useful for comparison, but they do not replace historical or ethnographic context. The most expressive final product would combine computational comparison with careful attribution to the source sky-culture traditions.

## ACKNOWLEDGEMENTS

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## REFERENCES

1. D. Bucur. The network signature of constellation line figures. *PLOS ONE* 17(7): e0272270, 2022.
2. K. Börner. *Atlas of Knowledge: Anyone Can Map*. MIT Press, 2015.
3. K. Börner, A. Bueckle, and M. Ginda. "Data visualization literacy: Definitions, conceptual frameworks, exercises, and assessments." *PNAS* 116(6): 1857-1864, 2018.